

Software Architect

1. Networking Foundations and Protocol Design

- **YouTube:**

- **Hussein Nasser's Network Engineering Playlist** – The industry baseline for understanding OSI layers, TCP/UDP sockets, TLS handshakes, multiplexing, and HTTP/1 vs HTTP/2 vs HTTP/3.

- **Ben Eater - Networking Tutorials** – Low-level visual breakdown of packet structures, routing tables, and how bits physically move across networks.

- **Hands-on Labs:**

- **Wireshark Packet Analysis Labs** – Capture live traffic on your system to manually inspect TCP flags, handshakes, and DNS requests.

- **Cisco Packet Tracer Labs** – Sandbox environment to simulate networks, configure routers, and model complex subnet topologies.

- **Documentation:**

- **MDN Web Docs: HTTP Network Protocols** – Clear, comprehensive specification of web network mechanics and standard protocol behaviors.

2. Relational Data Modeling and ACID Compliance

- **YouTube:**

- **Hussein Nasser - Database Engineering (ACID playlist)** – Master-class deep dives explaining write-ahead logging (WAL), transaction isolation levels, page formats, and locking mechanisms.

- **Pragmatic System Design - Database Indexing Internals** – Clear breakdowns detailing exactly how B-Trees and LSM-Trees navigate raw storage disk blocks.

- **Hands-on Labs:**

- **Hackattic - PostgreSQL Sandbox Challenges** – Real-world database troubleshooting labs focusing on resolving deadlocks, connection leaks, and index degradation.

- **Use The Index, Luke! Interactive Guides** – Highly practical hands-on templates mapping query structure adjustments directly to database performance gains.

- **Documentation:**

- **PostgreSQL Official Server Administration Manual** – The ultimate standard for understanding transaction semantics, write-ahead logs, and database

internals.

3. API Design, Contracts, and Integration Patterns

- **YouTube:**

- [ByteByteGo by Alex Xu - API Design Trends](#) – Visually contrasts REST, GraphQL, gRPC, and WebSockets while highlighting payload serialization efficiencies.
- [Tech Primers - Protocol Buffers and gRPC](#) – Step-by-step technical guides explaining binary serialization over HTTP/2.

- **Hands-on Labs:**

- [Postman API Network Labs](#) – Live interactive platform practicing API contract design, mock servers, automated testing suites, and OpenAPI/Swagger specifications.
- [Buf Schema Registry Quickstarts](#) – Coding exercises focused on building, linting, and managing proto files for contract-driven APIs.

- **Documentation:**

- [Google API Design Guide](#) – The gold-standard architectural manual outlining strict naming conventions, error mappings, and resource patterns for web microservices.

4. Distributed Data Systems and CAP Trade-offs

- **YouTube:**

- [MIT 6.824: Distributed Systems](#) – Legendary, mathematically rigorous university lecture series detailing consensus algorithms (Raft, Paxos), sharding, replication loops, and fault boundaries.
- [Martin Kleppmann - Distributed Systems Lecture Series](#) – Visually rich, foundational theoretical walkthroughs tracking vector clocks, linearizability, and eventual consistency.

- **Hands-on Labs:**

- [Fly.io - Distributed Systems Challenges \(Maelstrom\)](#) – Test real distributed node networks (implementing gossip protocols and replicate state machines) directly on your machine.
- [AWS Workshop Studio \(DynamoDB Scaling Paths\)](#) – Sandbox profiles modeling partitioned datasets and evaluating CAP theorem behavior during infrastructure drops.

- **Documentation:**

- [Designing Data-Intensive Applications \(DDIA Book Reference\)](#) – The ultimate industry Bible mapping partitioning strategy, replication lags, and transactions in

distributed clusters.

5. Event-Driven Architecture and Messaging Infrastructure

- **YouTube:**

- [Confluent Kafka Engineering Channel](#) – Incredible visual guides covering log files, consumer groups, message offset math, and exactly-once delivery guarantees.

- [Tech Primers - RabbitMQ vs Kafka Mechanics](#) – Clear architectural breakdowns contrasting smart-broker queue systems against dumb-broker append-only logs.

- **Hands-on Labs:**

- [RabbitMQ Official Developer Tutorials](#) – Implement exact code patterns for task queues, routing exchanges, publisher confirms, and dead-letter drops.

- [Killercoda Event Streaming Playgrounds](#) – Spin up instant, in-browser multi-broker Kafka configurations to practice topic distribution logic.

- **Documentation:**

- [Apache Kafka Core Documentation](#) – Production framework outlining storage tier parameters, retention laws, and consumer rebalancing architectures.

6. Caching, Performance Tuning, and Gateways

- **YouTube:**

- [Hussein Nasser - Proxies, Reverse Proxies, and Load Balancers](#) – Comprehensive look at Layer 4 vs Layer 7 traffic routing, connection pooling, and edge proxies like NGINX and Envoy.

- [Redis Labs Engineering Videos](#) – Explains caching topologies, eviction mechanisms (LRU/LFU), and state strategies like Cache-Aside vs Write-Through.

- **Hands-on Labs:**

- [Killercoda Redis Scenarios](#) – In-browser sandbox terminals configuring data structures, clustering, and persistence configurations.

- [Envoy Proxy Sandbox Setup Guides](#) – Use docker-compose configurations to manually run rate-limiting loops and circuit-breaking controls.

- **Documentation:**

- [Redis Documentation: Caching Patterns & NGINX Performance Tuning Guide](#) – Production blueprints for optimizing high-throughput traffic endpoints.

7. Microservices Decomposition, CQRS, and Event Sourcing

- **YouTube:**

- [Amichai Mantinband - Clean Architecture & DDD](#) – Outstanding visual breakdowns of Domain-Driven Design boundaries, aggregate roots, CQRS read-

sides, and transaction patterns.

- Microservices Patterns by Chris Richardson – Complete structural walk-through detailing Saga Patterns, Transactional Outboxes, and service mapping traps.

- **Hands-on Labs:**

- Microsoft eShopOnContainers Microservices Blueprint – The baseline open-source repository used to run, explore, and analyze multi-container microservice patterns locally.

- Dapr (Distributed Application Runtime) Quickstarts – Practical coding tutorials modeling state abstraction, service invocations, and publish-subscribe sidecars.

- **Documentation:**

- Microservices.io Architectural Pattern Catalog – The definitive structural dictionary mapping prescriptive software solutions to distributed computing flaws.

8. Containerization, Kubernetes, and Cloud Runtimes

- **YouTube:**

- TechWorld with Nana - Kubernetes Roadmap – The gold-standard overview detailing multi-node clusters, control planes, network overlays, pods, and deployments.

- Programming with Mosh - Docker Deep Dive – Flawless execution guides on layer optimization, micro-sizing base images, and secure volume management.

- **Hands-on Labs:**

- Play with Docker & Play with Kubernetes – Free, browser-based sandboxes allowing you to configure multi-node clusters instantly.

- Killercoda Kubernetes Tasks – Direct operational drills challenging you to troubleshoot failing control planes and invalid YAML definitions.

- **Documentation:**

- Kubernetes Official Tasks Reference Manual – The authoritative blueprint guide used by platform engineers to configure production cluster environments.

9. Application Security, Cryptography, and Identity

- **YouTube:**

- OktaDev / Aaron Parecki - OAuth 2.0 & OIDC Deep Dive – The absolute step-by-step master class on authorization flows, token issuance, verification, and PKCE extensions.

- Pringle Cyber - Cryptographic Keys and JWT Breaches – Explains symmetric vs asymmetric encryption, hashing salts, and token signature verification risks.

- **Hands-on Labs:**

- **PortSwigger Web Security Academy** – The world's top playground for safely executing attacks against broken authentication, OAuth flaws, and broken cryptography.

- **OWASP Juice Shop Sandbox Application** – Run a vulnerable microservice local container to practice discovering token leakage and fixing data configurations.

- **Documentation:**

- **OWASP Top 10 Security Verification Standards & OAuth 2.0 Official Specs** – Enterprise-grade security verification checklists used to secure perimeter APIs.

10. System Resilience, Reliability, and Observability

- **YouTube:**

- **Grafana & Prometheus Setup by TechWorld with Nana** – Clear architectural explanation of pull-based time-series metrics, scraping, and dashboard design.

- **Datadog Community Engineering Guides** – Demonstrates distributed tracing (APM tracking) across multiple asynchronous API hops.

- **Hands-on Labs:**

- **Grafana Play Workspace** – Live dashboard exploration platform demonstrating query builders, log parsing, and altering frameworks.

- **Chaos Mesh Interactive Scenarios** – Practice injecting network latency, disk pressure, and random pod deletions directly into active software runtimes.

- **Documentation:**

- Google SRE Book - Monitoring & Alerting Principles – The foundational industry text establishing standard metrics structures for SLOs, SLAs, and error budgets.

11. Enterprise Architecture, Governance, and Scaled Delivery

- **YouTube:**

- **GOTO Conferences - Software Architecture Track** – Curated presentations from principal and fellow tech engineers detailing system code retirement, tech debt curation, and structural migrations.

- **Continuous Delivery Channel by Dave Farley** – High-yield industry insights explaining how to build deployment safety gates without sacrificing velocity.

- **Hands-on Labs:**

- **Architecture Katas Challenge Workspace** – Real architectural prompt generators that challenge you to design scalable enterprise systems within strict budget and timeline constraints.

- **SonarQube Quality Gate Integrations** – Run automated code metrics analysis setups locally to build and enforce strict quality standard templates.
- **Documentation:**
 - **The Pragmatic Programmer Core Framework Guides** – Essential architectural governance strategies targeting highly reliable, stable enterprise software delivery.